Proceedings of the International Symposium on Soil Health and Sustainable Crop Production

Volume 1

Edited by

Rotimi Alade

A demonstrative volume prepared as a LaTeX benchmark asset.

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Nature of this volume. This volume is a demonstrative publication prepared as a LaTeX benchmark asset. The symposium is fictional. The contributed papers, their field trials, sites and datasets are synthetic: every measurement is generated by a seeded computer program supplied with the asset, and no value reports an experiment carried out on real land. The papers cite real published literature for methods, definitions and context; those works are cited, not reproduced. The results in this volume must not be used as evidence about real soils, crops or farming systems.

Conventions. Quantities are given in SI units. Grain and biomass yields are expressed in $\mathrm{t\,ha^{-1}}$ on a dry-matter basis unless stated otherwise, soil organic carbon in $\mathrm{g\,kg^{-1}}$, and soil pH is measured in water at a soil-to-solution ratio of 1:2.5.

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Editor's Preface

Soil is the foundation of crop production, yet its condition changes slowly and is easily overlooked until yields begin to fall. The idea of soil health treats the soil as a living system whose chemical, physical and biological properties together decide how well it supports crops, stores water and carbon, and recovers from disturbance. Measuring that condition, and managing land so that it improves rather than declines, is a shared concern of agronomists, soil scientists and farmers in every climatic zone.

This volume brings together six contributed papers presented at the International Symposium on Soil Health and Sustainable Crop Production. They were chosen to span the main questions of the field: how soil health is indicated and measured, how nutrients and organic matter are managed, how crops respond to different systems of tillage, intercropping and rotation, and how results from field trials are analysed and reported. Each paper is self-contained, with its own abstract and list of references, and each can be read and typeset on its own.

All trial data in this volume are synthetic, as the imprint explains. They were designed to behave like realistic field measurements and to obey the rules set out in the methods of each paper, so that the volume can serve as a faithful example of how such work is written, checked and published.

Rotimi Alade
Editor

How This Volume Is Organised

Structure

The volume is divided into three parts. The front matter contains the title pages, the contents, the editor's preface and this note. The main matter contains the six contributed papers, one chapter each. The back matter contains the list of abbreviations and the subject index.

The papers

Each paper opens with its title, a short title used in the running head, and an abstract. It then follows the usual order of a research paper: introduction, materials and methods, results, discussion and conclusion. References are listed at the end of each paper rather than collected at the end of the volume, so that every paper can be cited and read on its own.

Data, tables and figures

The data behind every table and figure are stored with the volume as tab-separated files. Figures are drawn directly from those files when the volume is typeset, and values quoted in the text are computed from the same data, so text, tables and figures cannot disagree with one another.

Abbreviations and index

Abbreviations are spelled out at their first use in each paper and are listed together in the back matter. The subject index points to the pages on which the main concepts, methods and measured properties are discussed.

Paper 1

Selecting a Minimum Dataset of Soil Health Indicators for Mixed Cropping Systems

Contributed paper

Abstract. Soil health cannot be measured directly; it has to be inferred from a small number of soil properties that respond to management and that relate to the functions a farmer depends on. This paper describes how a minimum dataset of six indicators was chosen for a multi-site trial network covering three climatic zones, how each indicator is converted to a common score between 0 and 1, and how the scores are combined into a single soil health index. The dataset spans chemical, physical and biological properties so that no single group of processes dominates the index. The approach is applied to synthetic trial data and is intended as a worked example of indicator selection and scoring, not as a recommendation for any real location.

Keywords: soil health; soil quality; minimum dataset; indicator scoring; soil health index

1.1 Introduction

Interest in the condition of agricultural soils has grown steadily since soil quality was defined as the capacity of a soil to function within ecosystem boundaries, sustaining biological productivity, environmental quality and plant and animal health (Karlen et al. 1997). The term *soil health* places more weight on the soil as a living system, and on the organisms whose activity drives nutrient cycling and the formation of soil structure (Doran and Zeiss 2000). The two terms are now often used interchangeably, although the concept of soil health has broadened to include links with human health and with the services that soils provide beyond the farm (Lehmann et al. 2020).

Because soil health is a property of the whole system, it cannot be read from a single instrument. It is inferred instead from indicators: measurable soil properties that are sensitive to management, related to a soil function, and practical to measure. The number of candidate indicators is large, and a review of assessment schemes found that the choice between them, and the way measured values are interpreted, is still inconsistent from one programme to the next (Bünemann et al. 2018). A common response is to select a *minimum dataset*, the smallest set of indicators that together describe the functions of interest without redundancy.

Selecting indicators is only the first of three steps. Each measured value must then be interpreted, usually by transforming it to a unitless score that expresses how well the soil performs relative to a target, and the scores must finally be combined into an index. The Soil Management Assessment Framework formalised these three steps and showed that scoring curves which rise, fall or peak with the measured value can represent the different ways in which an indicator relates to soil function (Andrews et al. 2004).

This paper applies the same three steps to the trial network used throughout this volume. Its aims are to state the criteria by which six indicators were chosen, to define the scoring function for each, and to show how the resulting index is calculated, so that the later papers can report soil health on a common basis. Soil organic carbon is determined by wet oxidation (Walkley and Black 1934) and extractable phosphorus by the Mehlich 3 method (Mehlich 1984); the remaining indicators and all laboratory procedures are described under Materials and Methods.

1.2 Materials and Methods

1.2.1 Trial and sampling

The indicators were measured in a trial comparing three management systems at each of the three sites of the trial network: a conventional system with tillage and mineral fertiliser only (CONV), a conservation system with reduced tillage and retained residues (CONS), and an integrated system combining reduced tillage, residues, organic inputs and fertiliser (INTG). The systems were arranged in randomised complete blocks with four blocks per site. Topsoil was sampled to a depth of 0.20 m from every plot. The data are held in `data/trials/t1_indicators.tsv`.

1.2.2 Selection of indicators

Candidate indicators were retained only if they met four criteria: they respond to management within a few seasons, they relate to at least one soil function, they can be measured at moderate cost, and they are not strongly redundant with another retained indicator (Andrews et al. 2004; Bünemann et al. 2018). Six indicators met the criteria: soil organic carbon (SOC), soil pH, extractable phosphorus (P), bulk density (BD), water-stable aggregates (WSA) and microbial biomass carbon (MBC). Together they cover chemical, physical and biological properties (Doran and Zeiss 2000). Their laboratory methods are listed in Table 1.1; SOC was determined by wet oxidation (Walkley and Black 1934) and P with the Mehlich 3 extractant (Mehlich 1984).

Table 1.1: Indicators of the minimum dataset, their laboratory methods and scoring parameters. For *more is better* indicators the score rises from 0 at the lower limit L to 1 at the upper limit U; for *less is better* it falls from 1 at L to 0 at U; for *optimum range* it is 1 between O_1 and O_2 and falls to 0 at L and U.

Indicator	Unit	Method	Curve	L	O_1	O_2	U
SOC	$\mathrm{g\,kg^{-1}}$	wet oxidation	more is better	5			25
pH	–	water, 1:2.5	optimum range	4.5	6.0	7.0	8.5
P	$\mathrm{mg\,kg^{-1}}$	Mehlich 3	more is better	5			40
BD	$\mathrm{g\,cm^{-3}}$	core method	less is better	1.10			1.60
WSA	%	wet sieving	more is better	20			70
MBC	$\mathrm{mg\,kg^{-1}}$	fumigation–extraction	more is better	50			300

1.2.3 Scoring

Each measured value x was converted to a score s between 0 and 1. For an indicator where more is better,

$$s = \min\!\left(1,\ \max\!\left(0,\ \frac{x - L}{U - L}\right)\right), \tag{1.1}$$

and for an indicator where less is better the score is $1 - s$ with the same limits. For soil pH, which has an optimum range from O_1 to O_2, the score is 1 inside the range and falls linearly to 0 at L below and U above:

$$s = \begin{cases} 0 & x \le L \text{ or } x \ge U, \\ \dfrac{x - L}{O_1 - L} & L < x < O_1, \\ 1 & O_1 \le x \le O_2, \\ \dfrac{U - x}{U - O_2} & O_2 < x < U. \end{cases} \tag{1.2}$$

The limits in Table 1.1 follow the shape of the scoring curves of the Soil Management Assessment Framework (Andrews et al. 2004); their values were set for this volume and are not calibrated for any real soil.

1.2.4 Soil health index

The soil health index (SHI) of a plot is the mean of its six indicator scores,

$$\mathrm{SHI} = \frac{1}{n}\sum_{i=1}^{n} s_i, \qquad n = 6, \tag{1.3}$$

so that every indicator carries equal weight and the index lies between 0 and 1. The index of each system was compared between systems within each site.

1.3 Results

The mean values of the six indicators are given in Table 1.2. The integrated system had the highest soil organic carbon, water-stable aggregates and microbial biomass carbon at every site, and the conventional system the lowest, while bulk density was lowest under the integrated system. Soil pH differed little between systems and remained close to the starting value of each site.

The soil health index ranged from 0.342 to 0.880 (Figure 1.1). At the temperate site S1 it rose from 0.682 under the conventional system to 0.796 under the conservation system and 0.880 under the integrated system. The same order held at the semi-arid site S2, where the index was lowest throughout (0.342, 0.426 and 0.502), and at the humid tropical site S3 (0.566, 0.707 and 0.790). The effect of the management system on the index was significant at each site (S1: $F = 179.38$, $P < 0.001$; S2: $F = 66.22$, $P < 0.001$; S3: $F = 102.67$, $P < 0.001$; 2 and 6 degrees of freedom).

1.4 Discussion

The index ranked the three systems in the same order at every site, which is the behaviour a soil health index needs if it is to guide management: the integrated system, which returns the most organic matter to the soil, scored highest, and the conventional system lowest. The ranking was

Table 1.2: Mean values of the six indicators and the soil health index (SHI) by site and management system (four blocks). SOC and MBC in $\mathrm{g\,kg^{-1}}$ and $\mathrm{mg\,kg^{-1}}$, P in $\mathrm{mg\,kg^{-1}}$, BD in $\mathrm{g\,cm^{-3}}$, WSA in per cent.

Site	System	SOC	pH	P	BD	WSA	MBC	SHI
S1	CONV	18.3	6.36	39.7	1.31	41.7	165	0.682
S1	CONS	19.8	6.46	40.3	1.28	52.7	238	0.796
S1	INTG	21.6	6.48	46.9	1.25	59.8	293	0.880
S2	CONV	7.5	7.60	28.6	1.47	34.4	78	0.342
S2	CONS	8.3	7.58	30.1	1.47	43.0	133	0.426
S2	INTG	9.0	7.81	35.4	1.42	49.3	186	0.502
S3	CONV	13.8	5.41	33.1	1.26	47.6	129	0.566
S3	CONS	16.0	5.44	38.0	1.23	57.7	212	0.707
S3	INTG	16.6	5.56	41.3	1.20	62.2	254	0.790

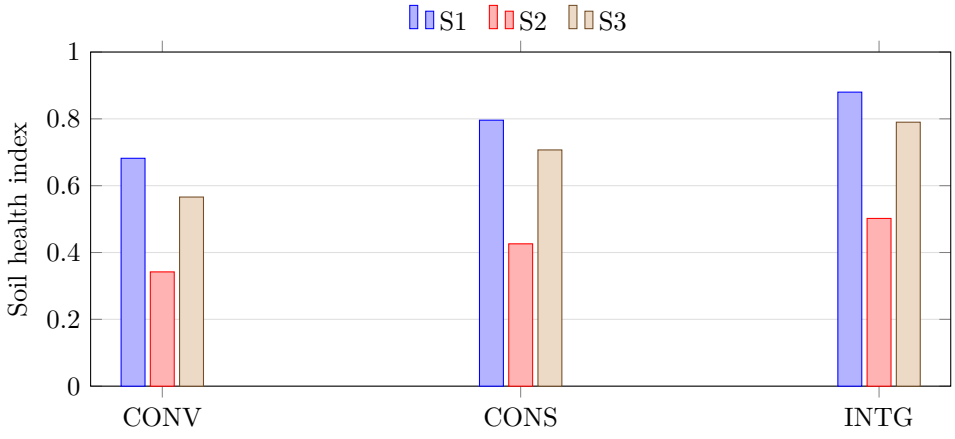


Figure 1.1: Mean soil health index of the three management systems at each site.

driven mainly by the biological and physical indicators. Soil organic carbon, aggregate stability and microbial biomass all respond to organic inputs and reduced disturbance, and their scores moved together, as the links between soil organisms, organic matter and structure would lead one to expect (Doran and Zeiss 2000).

The index also separated the sites more strongly than the systems. The semi-arid site scored lowest under every system, because its low carbon and high bulk density kept several indicators near the bottom of their scoring range. This illustrates a known limitation of indices built on fixed scoring limits: they mix the inherent capacity of a soil with the effect of management, so a sandy soil in a dry climate can appear unhealthy even when it is well managed (Bünemann et al. 2018). Comparing systems within a site, as done here, avoids that confusion; comparing sites requires scoring limits that reflect the potential of each soil, as the Soil Management Assessment Framework proposes (Andrews et al. 2004).

Two further cautions apply. Equal weighting assumes that every indicator is equally important to the functions of interest, which is a choice rather than a finding. And soil pH, scored on an optimum range, contributed little to the differences between systems because it changed little within the duration of the trial; its role in the index is to flag soils that move out of the range in which nutrients are available.

1.5 Conclusion

A minimum dataset of six indicators, converted to scores and averaged into a single index, ranked three management systems consistently at three contrasting sites, with the index ranging from 0.342 to 0.880. The index is best used to compare management within a site; comparison across sites requires scoring limits matched to each soil. All values in this paper come from synthetic data and illustrate the method only.

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Paper 2

Maize Yield Response to Nitrogen Rate and Nitrogen Use Efficiency

Contributed paper

Abstract. Nitrogen is the nutrient that most often limits cereal yields, yet a large share of the nitrogen applied as fertiliser is not taken up by the crop. This paper reports a nitrogen rate trial in maize with five rates of mineral nitrogen, repeated at three sites in contrasting climatic zones. Grain yield is related to the rate applied through a response function, from which an agronomic optimum rate is derived, and the efficiency with which applied nitrogen is converted to grain is expressed through standard efficiency measures calculated against an unfertilised control. The trial data are synthetic and serve to illustrate how rate response and nitrogen use efficiency are calculated and reported.

Keywords: maize; nitrogen fertiliser; rate response; agronomic efficiency; nitrogen use efficiency

2.1 Introduction

Cereal production depends heavily on fertiliser nitrogen, and the efficiency with which that nitrogen is used has become a central question for both farm profitability and the environment. Reviews of cereal systems have shown that the fraction of applied nitrogen recovered by the crop is often low, and that the nitrogen not taken up is at risk of loss to water and air (Cassman et al. 2002). Improving efficiency therefore depends on matching the supply of nitrogen, in amount and timing, to the demand of the crop.

A retrospective analysis of fertiliser nitrogen in cereal production found that recovery efficiencies measured in farmers' fields are commonly lower than those achieved on research stations, and that the gap reflects both management and the variable nitrogen supply of the soil itself (Ladha et al. 2005). Nitrogen use efficiency is not a single number: it can be expressed as the extra grain produced per unit of nitrogen applied, as the fraction of applied nitrogen taken up by the crop, or as the grain produced per unit of nitrogen taken up, and each measure answers a different question.

Rate trials remain the basic tool for describing how a crop responds to nitrogen. Yield rises steeply at low rates, more slowly at intermediate rates, and reaches a plateau once nitrogen no longer limits growth. Fitting a response function to the yields allows the rate at which the marginal return falls to zero to be estimated, and comparison with an unfertilised control allows the efficiency measures to be calculated. Such trials are usually laid out as randomised complete blocks, which control for gradients in soil fertility across the field (Gomez and Gomez 1984).

Fertiliser is most effective when it is used together with organic inputs and well-adapted varieties, an approach set out as integrated soil fertility management (Vanlauwe et al. 2010). This paper reports the mineral nitrogen component of such a system. Its aims are to describe the yield response of maize to five rates of nitrogen at three sites, to estimate the agronomic optimum rate at each site, and to report the agronomic efficiency of applied nitrogen across rates and sites.

2.2 Materials and Methods

2.2.1 Treatments and design

Maize received five rates of mineral nitrogen, 0, 40, 80, 120 and 160 kg ha^{-1}, applied as urea in two equal splits at sowing and at the six-leaf stage. Phosphorus and potassium were applied to all plots at rates that did not limit growth. The rates were arranged in randomised complete blocks with four blocks at each of the three sites (Gomez and Gomez 1984). Grain yield is expressed in t ha^{-1} on a dry-matter basis, and nitrogen uptake as the nitrogen in grain and stover at harvest. The data are held in `data/trials/t2_nitrogen.tsv`.

2.2.2 Yield response

The response of grain yield Y to the nitrogen rate N was described by a quadratic-plateau function,

$$Y = \begin{cases} a + bN + cN^2 & N < N_p, \\ Y_p & N \geq N_p, \end{cases} \qquad N_p = -\frac{b}{2c}, \tag{2.1}$$

where a, b and c are fitted coefficients ($c < 0$), N_p is the rate at which the response reaches its plateau and Y_p is the plateau yield. N_p is taken as the agronomic optimum rate.

2.2.3 Nitrogen use efficiency

Two efficiency measures were calculated for each fertilised plot against the unfertilised plot of the same block (Cassman et al. 2002; Ladha et al. 2005). The agronomic efficiency of applied nitrogen, in kilograms of grain per kilogram of nitrogen, is

$$\mathrm{AE}_N = \frac{1000\,(Y_N - Y_0)}{N}, \tag{2.2}$$

where Y_N and Y_0 are the grain yields (t ha^{-1}) with and without nitrogen. The apparent recovery efficiency, the fraction of the applied nitrogen taken up by the crop, is

$$\mathrm{RE}_N = \frac{U_N - U_0}{N}, \tag{2.3}$$

where U_N and U_0 are the nitrogen uptakes (kg ha^{-1}) with and without nitrogen.

2.3 Results

2.3.1 Yield response

Grain yield increased with the nitrogen rate at all three sites (Table 2.1). Without nitrogen, mean yield was 4.73, 1.82 and 2.48 t ha^{-1} at S1, S2 and S3; at 120 kg ha^{-1} it reached 9.57, 5.68 and 7.59 t ha^{-1}, and the highest rate added little or nothing further.

The quadratic-plateau function described the response well, with coefficients of determination of 0.976, 0.931 and 0.982 (Figure 2.1). The agronomic optimum rate was 143 kg ha^{-1} at S1, 125 kg ha^{-1} at S2 and 137 kg ha^{-1} at S3, with plateau yields of 9.83, 5.69 and 7.72 t ha^{-1}.

Table 2.1: Mean grain yield, nitrogen uptake, agronomic efficiency (AE_N, kg grain per kg N) and apparent recovery efficiency (RE_N) by site and nitrogen rate (four blocks). Efficiencies are not defined for the unfertilised control.

Site	N (kg ha^{-1})	Yield (t ha^{-1})	Uptake (kg ha^{-1})	AE_N	RE_N
S1	0	4.73	46.3	–	–
S1	40	6.98	73.3	56.2	0.67
S1	80	9.07	101.1	54.3	0.69
S1	120	9.57	108.4	40.4	0.52
S1	160	9.78	116.0	31.6	0.44
S2	0	1.82	20.7	–	–
S2	40	3.95	45.0	53.0	0.61
S2	80	5.18	62.1	41.9	0.52
S2	120	5.68	71.5	32.1	0.42
S2	160	5.38	75.3	22.2	0.34
S3	0	2.48	23.3	–	–
S3	40	4.83	52.1	58.8	0.72
S3	80	7.01	79.7	56.7	0.70
S3	120	7.59	91.8	42.6	0.57
S3	160	7.55	94.5	31.7	0.45

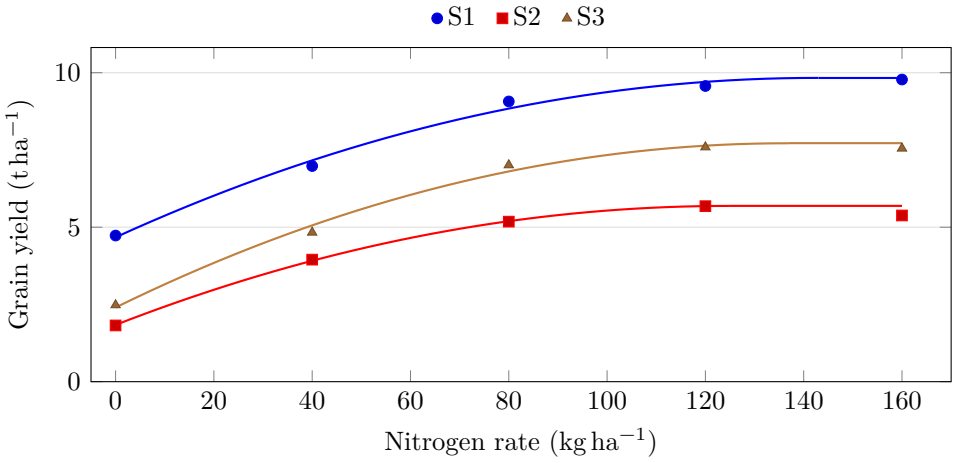


Figure 2.1: Mean grain yield of maize at five nitrogen rates (points) and the fitted quadratic-plateau response at each site (lines).

2.3.2 Nitrogen use efficiency

Agronomic efficiency was highest at the lowest rate and declined as the rate increased. At 40 kg ha^{-1} each kilogram of nitrogen produced 56.2, 53.0 and 58.8 kg of additional grain at S1, S2 and S3; at 160 kg ha^{-1} the corresponding values fell to 31.6, 22.2 and 31.7 kg. The apparent recovery of nitrogen followed the same pattern (Table 2.1).

2.4 Discussion

The response of maize to nitrogen followed the familiar pattern of diminishing returns, and the optimum rate differed between sites in line with their yield potential: it was highest at the temperate site, where water was least limiting, and lowest at the semi-arid site, where yield was capped by water rather than by nitrogen. A single recommended rate across the three sites would therefore have over-fertilised the dry site and under-fertilised the temperate one, which is why site-specific recommendations are central to improving nitrogen use efficiency (Cassman et al. 2002).

Agronomic and recovery efficiency both fell as the rate increased. This is expected, because each additional kilogram of nitrogen is applied to a crop whose demand is already closer to being met. The practical consequence is a trade-off: rates near the agronomic optimum maximise yield, but the last increments of nitrogen are used least efficiently and contribute most to the surplus at risk of loss. The recovery values found here lie within the range reported for cereal systems, where much of the applied nitrogen is not taken up by the crop (Ladha et al. 2005).

The optimum estimated from a quadratic-plateau function is sensitive to the form of the function and to the spread of rates tested, and it ignores the prices of grain and fertiliser. An economic optimum, at which the value of the extra grain equals the cost of the extra nitrogen, is always lower than the agronomic optimum. Combining moderate rates of mineral nitrogen with organic inputs, as integrated soil fertility management recommends, is one way to retain most of the yield benefit while improving efficiency (Vanlauwe et al. 2010).

2.5 Conclusion

Maize yield rose with nitrogen to a plateau at all three sites, with agronomic optimum rates of 143, 125 and 137 kg ha^{-1} at S1, S2 and S3. Nitrogen use efficiency declined steadily with rate, so the highest rates gave little extra grain for the nitrogen applied. Recommendations should be set site by site and weighed against efficiency as well as yield. The data are synthetic and the values illustrate the calculation only.

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Paper 3

Maize–Cowpea Intercropping and the Land Equivalent Ratio

Contributed paper

Abstract. Growing a cereal and a legume together on the same land can produce more in total than growing each crop alone on the same area. This paper compares maize and cowpea grown as sole crops with the two crops grown together in alternating rows, at three sites. The advantage of the intercrop is measured by the land equivalent ratio, calculated from the intercrop yield of each species relative to its sole-crop yield, and the contribution of each crop to the combined advantage is reported through its partial ratio. All yields are synthetic; the paper demonstrates how the land equivalent ratio is calculated, tabulated and interpreted.

Keywords: intercropping; maize; cowpea; land equivalent ratio; cereal–legume systems

3.1 Introduction

Intercropping, the growing of two or more crops together for at least part of their life cycle, is an old practice that remains widespread in low-input farming systems. It has attracted renewed interest because it can raise total production without additional inputs, or make production more stable, and so offers one route towards the sustainable intensification of agriculture (Brooker et al. 2015). The advantage arises when the component crops use resources in complementary ways, for example by rooting at different depths, by drawing on light at different times, or, in the case of a legume, by fixing atmospheric nitrogen that the cereal would otherwise have to take from the soil.

Comparing an intercrop with sole crops is not straightforward, because the intercrop produces two harvests while each sole crop produces one. The land equivalent ratio resolves this by asking how much land would be needed under sole cropping to produce the yields obtained from one unit of land under intercropping. It is the sum, over the component crops, of the intercrop yield of each crop divided by its sole-crop yield (Mead and Willey 1980). A ratio greater than one shows that the intercrop uses land more efficiently than the sole crops; a ratio of 1.2, for example, means that 20 per cent more land would be needed to match the intercrop by growing the crops separately.

The value of the ratio depends on the sole-crop yields used to standardise it, and the choice of those yields must be stated when results are reported (Mead and Willey 1980). In this trial each sole crop is grown in the same blocks as the intercrop, so that the standardising yields

come from the same site, season and management. Cereal–legume combinations are also a component of integrated soil fertility management, in which legumes contribute to the nitrogen economy of the following cereal (Vanlauwe et al. 2010).

The aims of this paper are to report the sole-crop and intercrop yields of maize and cowpea at three sites, to calculate the partial and total land equivalent ratios, and to identify the sites at which the intercrop gives a yield advantage.

3.2 Materials and Methods

3.2.1 Treatments and design

Three cropping systems were compared: sole maize (SOLE_M), sole cowpea (SOLE_C), and an intercrop of the two crops in alternating single rows (INTER). Sole crops were sown at their recommended densities; in the intercrop each crop occupied half of the rows. The systems were arranged in randomised complete blocks with four blocks at each of the three sites (Gomez and Gomez 1984), so that every intercrop plot has sole-crop plots of both species in the same block. Yields are expressed in $\mathrm{t\,ha^{-1}}$ of dry grain. The data are held in `data/trials/t3_intercrop.tsv`; a sole crop has no yield for the other species, recorded as `NA`.

3.2.2 Land equivalent ratio

For each block the partial land equivalent ratios of maize and cowpea are

$$\mathrm{LER}_m = \frac{Y_{im}}{Y_{sm}}, \qquad \mathrm{LER}_c = \frac{Y_{ic}}{Y_{sc}}, \tag{3.1}$$

where Y_{im} and Y_{ic} are the intercrop yields of maize and cowpea and Y_{sm} and Y_{sc} their sole-crop yields in the same block. The total land equivalent ratio is their sum (Mead and Willey 1980),

$$\mathrm{LER} = \mathrm{LER}_m + \mathrm{LER}_c. \tag{3.2}$$

Standardising each intercrop by the sole crops of its own block keeps the comparison within the same site, season and soil. An intercrop is said to give a yield advantage at a site when the mean LER over the four blocks exceeds 1.

3.3 Results

In every block, each crop yielded less in the intercrop than as a sole crop, as expected when each species occupies only half of the rows. The combined production of the intercrop, however, exceeded that of the sole crops on an equal-area basis at all three sites (Table **??**).

Table 3.1: Mean partial land equivalent ratios of maize and cowpea, mean total land equivalent ratio (LER), and the lowest and highest block values of the total ratio at each site (four blocks).

Site	LER_m	LER_c	LER	Lowest	Highest
S1	0.73	0.49	1.23	1.15	1.31
S2	0.68	0.60	1.28	1.22	1.35
S3	0.81	0.56	1.37	1.32	1.40

The mean total land equivalent ratio was 1.23 at S1, 1.28 at S2 and 1.37 at S3: to match one hectare of intercrop, sole crops would have needed between 1.23 and 1.37 hectares of land. Maize contributed the larger share at every site, with partial ratios of 0.73, 0.68 and 0.81; the

partial ratios of cowpea were 0.49, 0.60 and 0.56 (Figure 3.1). The highest ratio was found at the humid tropical site, where maize retained the largest fraction of its sole-crop yield in the intercrop.

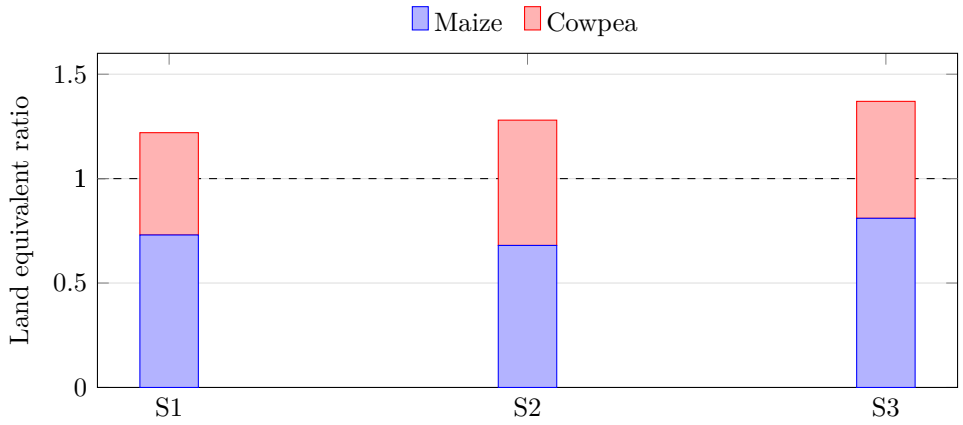


Figure 3.1: Mean partial land equivalent ratios of maize and cowpea at each site. The height of each bar is the total ratio; the dashed line marks a ratio of 1.

3.4 Discussion

A total land equivalent ratio above one at every site shows that the two crops used the available resources more completely together than apart. The partial ratios indicate how the advantage arose. Maize, the taller and more competitive crop, retained most of its sole-crop yield, while cowpea lost a larger share of its yield to shading and competition yet still contributed enough to lift the total well above one. This asymmetry is typical of cereal–legume mixtures, in which complementarity in the use of light, water and nitrogen allows the combined yield to exceed what either crop could produce on the same land alone (Brooker et al. 2015).

The advantage was largest at the humid tropical site and smallest at the temperate site. Where the growing season is long and water is not limiting, the two crops can occupy the canopy for longer and use more of the resources that a single crop would leave unused. The comparison must nonetheless be read with care: the land equivalent ratio depends on the sole-crop yields used to standardise it, and on the relative proportions of the two crops in the intercrop (Mead and Willey 1980). Both were held constant here, so the ratios can be compared between sites but not with trials that used other row arrangements.

The land equivalent ratio measures the efficiency of land use, not the value of the harvest. Because maize and cowpea grain differ in price and in their role in the diet, a farmer may prefer an arrangement with a slightly lower ratio but a larger share of the more valuable crop. The nitrogen that cowpea leaves in the soil, a benefit to the following cereal (Vanlauwe et al. 2010), is also not captured by a ratio calculated within a single season.

3.5 Conclusion

Maize–cowpea intercropping gave a yield advantage at all three sites, with mean land equivalent ratios of 1.23, 1.28 and 1.32. Maize contributed the larger share of the advantage at every site. The ratio should be interpreted together with the value of each crop and the effect of the legume on the following season. The yields are synthetic and illustrate the calculation only.

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Paper 4

Tillage and Residue Management Effects on Soil Organic Carbon and Yield

Contributed paper

Abstract. Reduced tillage and the retention of crop residues are promoted to protect soils and build soil organic carbon, but their effect on crop yield varies with climate and management. This paper reports a factorial trial combining two tillage systems, conventional and no-till, with two residue treatments, removed and retained, at three sites. Soil organic carbon in the topsoil is compared with its value at the start of the trial, and grain yield is compared across the four combinations. The data are synthetic; the paper shows how a two-factor trial is analysed and how changes in soil organic carbon are reported alongside yield.

Keywords: conservation agriculture; no-till; crop residues; soil organic carbon; factorial trial

4.1 Introduction

Soil organic carbon lies at the centre of soil health. It supports soil structure, water retention and nutrient supply, and agricultural soils that have lost carbon through cultivation have a large capacity to regain it. Practices that increase the soil carbon pool include reduced tillage, the retention of residues, cover crops and the addition of manures, and the resulting gains in soil carbon can support higher crop yields on degraded land (Lal 2004).

Conservation agriculture combines three principles: minimal soil disturbance, permanent soil cover and crop diversification. A global meta-analysis of paired yield observations found that no-till on its own tends to reduce yields relative to conventional tillage, but that the reduction is smallest when no-till is combined with residue retention and crop rotation, and that the combination can raise rainfed yields in dry climates (Pittelkow et al. 2015). The yield effect of reduced tillage therefore depends on whether the other principles are applied with it.

Whether these principles can be applied in practice is a separate question. In smallholder farming systems, crop residues are often needed as livestock feed, weeding without tillage can increase labour, and the evidence for benefits has been argued to be less consistent than its promotion suggests (Giller et al. 2009). Trials that separate the effect of tillage from the effect of residues help to show which part of the package produces which effect.

This paper reports such a trial. Soil organic carbon is determined by wet oxidation (Walkley and Black 1934). The aims are to estimate the separate and combined effects of tillage and residue management on topsoil organic carbon and on grain yield, and to test whether the effect of tillage depends on the residue treatment.

4.2 Materials and Methods

4.2.1 Treatments and design

Two tillage systems, conventional tillage (CT) and no-till (NT), were combined factorially with two residue treatments, residues removed (R0) and residues retained on the surface (R1), giving four combinations. The combinations were arranged in randomised complete blocks with four blocks at each of the three sites and maintained for five years (Gomez and Gomez 1984). The data are held in `data/trials/t4_tillage.tsv`.

4.2.2 Measurements

Topsoil to 0.20 m was sampled at the start and at the end of the trial, and soil organic carbon was determined by wet oxidation (Walkley and Black 1934). The change in soil organic carbon of a plot is

$$\Delta \mathrm{SOC} = \mathrm{SOC}_{\mathrm{final}} - \mathrm{SOC}_{\mathrm{initial}} \quad (\mathrm{g\,kg^{-1}}). \tag{4.1}$$

Maize grain yield in the final year is expressed in $\mathrm{t\,ha^{-1}}$ on a dry-matter basis.

4.2.3 Statistical model

At each site, the response y_{ijk} of tillage i and residue treatment j in block k is described by

$$y_{ijk} = \mu + \beta_k + T_i + R_j + (TR)_{ij} + \varepsilon_{ijk}, \tag{4.2}$$

where T_i and R_j are the main effects of tillage and residues and $(TR)_{ij}$ their interaction. A significant interaction indicates that the effect of tillage depends on whether residues are retained.

4.3 Results

4.3.1 Soil organic carbon

Over five years, topsoil organic carbon declined under conventional tillage with residues removed and increased under every other combination (Table 4.1, Figure 4.1). The largest gains were under no-till with residues retained: 1.50, 0.62 and 1.10 $\mathrm{g\,kg^{-1}}$ at S1, S2 and S3, against -0.35, -0.20 and -0.42 $\mathrm{g\,kg^{-1}}$ under conventional tillage with residues removed. Both main effects were significant at every site (Table 4.2). The interaction was significant at S1 ($F = 5.83$, $P = 0.039$) and S3 ($F = 5.61$, $P = 0.042$), where no-till gained more carbon when residues were retained, but not at the semi-arid site S2 ($F = 0.20$, $P = 0.662$).

4.3.2 Grain yield

Yield responded differently from carbon. At the temperate site S1, no-till reduced yield ($F = 6.94$, $P = 0.027$), most clearly when residues were removed (7.14 against 8.27 $\mathrm{t\,ha^{-1}}$). At the semi-arid site S2 the effect of tillage depended on residues (interaction $F = 6.41$, $P = 0.032$): no-till with residues retained gave the highest yield of the four combinations, 4.88 $\mathrm{t\,ha^{-1}}$, and

Table 4.1: Mean change in soil organic carbon (ΔSOC, g kg^{-1}) over five years and final-year grain yield (t ha^{-1}) by site, tillage (CT conventional, NT no-till) and residue treatment (R0 removed, R1 retained); four blocks.

Site	Tillage	Residue	ΔSOC	Yield
S1	CT	R0	-0.35	8.27
S1	CT	R1	0.33	8.29
S1	NT	R0	0.25	7.14
S1	NT	R1	1.50	8.05
S2	CT	R0	-0.20	4.48
S2	CT	R1	0.25	4.62
S2	NT	R0	0.12	4.13
S2	NT	R1	0.62	4.88
S3	CT	R0	-0.42	6.26
S3	CT	R1	0.22	6.59
S3	NT	R0	0.08	6.34
S3	NT	R1	1.10	6.12

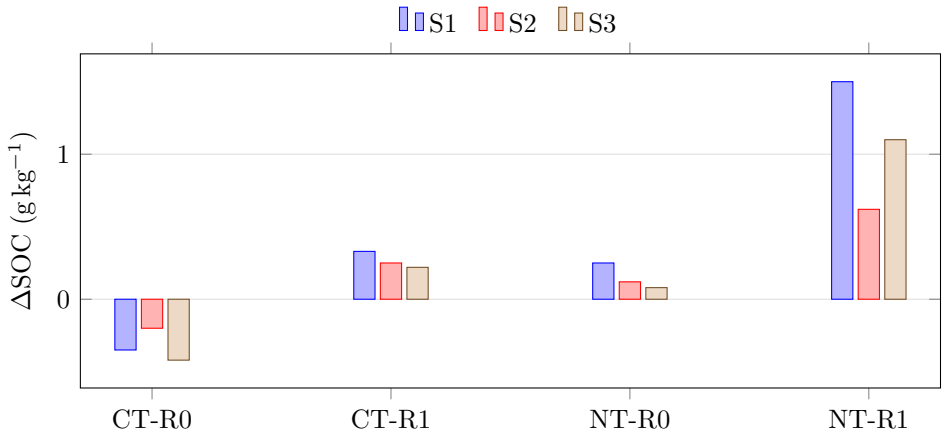


Figure 4.1: Mean change in topsoil organic carbon over five years under the four combinations of tillage and residue management at each site.

no-till with residues removed the lowest, 4.13 t ha^{-1}. At the humid tropical site S3 neither factor affected yield significantly (tillage $P = 0.374$; residues $P = 0.796$).

Table 4.2: Analysis of variance of the change in soil organic carbon (dsoc) and of grain yield at each site: F ratios and P values for the main effects of tillage and residues and their interaction. Each F has 1 and 9 degrees of freedom.

Site	Response	Source	F	P
S1	dsoc	tillage	55.53	<0.001
S1	dsoc	residue	65.31	<0.001
S1	dsoc	interaction	5.83	0.039
S1	yield	tillage	6.94	0.027
S1	yield	residue	3.19	0.108
S1	yield	interaction	2.99	0.118
S2	dsoc	tillage	40.09	<0.001
S2	dsoc	residue	73.84	<0.001
S2	dsoc	interaction	0.20	0.662
S2	yield	tillage	0.13	0.730
S2	yield	residue	13.03	0.006
S2	yield	interaction	6.41	0.032
S3	dsoc	tillage	75.42	<0.001
S3	dsoc	residue	111.91	<0.001
S3	dsoc	interaction	5.61	0.042
S3	yield	tillage	0.87	0.374
S3	yield	residue	0.07	0.796
S3	yield	interaction	1.78	0.215

4.4 Discussion

Both reduced tillage and residue retention increased topsoil organic carbon, and their combination gave the largest gain at every site. Residues supply the carbon input; reduced disturbance slows the breakdown of aggregates that protect organic matter from decomposition. Losses under conventional tillage with residues removed show that, without either, cultivated soils continue to lose carbon, consistent with the large carbon deficit of many agricultural soils (Lal 2004). Two cautions apply to these gains. They were measured in the topsoil only, and no-till tends to concentrate carbon near the surface, so part of the apparent gain may be a redistribution rather than a net increase in the whole profile.

The yield results differed from the carbon results and from site to site. At the temperate site, no-till lowered yield, particularly when residues were removed; at the semi-arid site, no-till with residues retained gave the highest yield; at the humid tropical site, neither factor mattered. This pattern mirrors the conclusion of a global meta-analysis, in which no-till reduced yields on average but performed best in dry climates and when combined with residue retention (**pittelkow2016**). Under dry conditions the residue mulch conserves soil water, which can outweigh the yield penalty of reduced tillage.

The results also show why conservation agriculture cannot be judged as a single package. In many smallholder systems, residues are fed to livestock and cannot simply be left on the field, and without residues the yield penalty of no-till may be largest (Giller et al. 2009). The factorial design used here separates the two components and shows that, at two of the three sites, the benefit of no-till depended on the residues being retained.

4.5 Conclusion

Over five years, no-till with residues retained raised topsoil organic carbon by 1.50, 0.62 and 1.10 $\mathrm{g\,kg^{-1}}$ at S1, S2 and S3, while the yield response to tillage depended on site and on residue management. Where residues cannot be retained, no-till alone is unlikely to deliver either the carbon or the yield benefit. All data are synthetic and illustrate the analysis only.

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Paper 5

Organic Amendments and the Stabilisation of Soil Organic Carbon

Contributed paper

Abstract. Adding organic materials to soil raises its carbon content, but only part of the added carbon remains in the soil for long. This paper compares three organic amendments, applied at equal rates of carbon, with an unamended control at three sites, and follows topsoil organic carbon over the course of the trial. The fraction of added carbon retained in the soil is calculated for each amendment, and the retention is related to soil texture through the clay content of each site. The data are synthetic and illustrate how carbon retention is calculated and how it can be interpreted in terms of the mechanisms that protect organic matter from decomposition.

Keywords: organic amendments; manure; compost; soil organic carbon; carbon retention; clay

5.1 Introduction

Organic amendments such as animal manure, compost and crop residues supply nutrients and add carbon to the soil. They are a core component of integrated soil fertility management, in which they are combined with fertiliser to raise the efficiency with which nutrients are used (Vanlauwe et al. 2010). Beyond their nutrient value they feed the soil organisms on which many soil functions depend (Doran and Zeiss 2000).

Most of the carbon in a fresh amendment is decomposed within a few seasons and returned to the atmosphere as carbon dioxide. The fraction that persists depends on how the remaining organic matter is protected. A review of the mechanisms concluded that the chemical resistance of organic compounds to decay is important only in the early stages of decomposition, and that long-term persistence depends instead on the physical inaccessibility of organic matter to decomposers and on its association with mineral surfaces (von Lützow et al. 2006).

Because organic matter is protected by association with silt and clay particles and by its occlusion within aggregates, soils differ in how much carbon they can stabilise, and the protective capacity of a soil sets an upper limit to the carbon it can hold in protected forms (Six et al. 2002). Soils with more clay would therefore be expected to retain a larger share of added carbon, while soils close to that upper limit would retain less of each new addition. These ideas have practical weight, because the carbon that agricultural soils could regain is considerable (Lal 2004).

The aims of this paper are to compare the gain in topsoil organic carbon under three amendments with that of an unamended control, to calculate the fraction of the added carbon retained under each amendment, and to examine whether retention is greater at sites with higher clay content.

5.2 Materials and Methods

5.2.1 Treatments and design

Three organic amendments, cattle manure (MANURE), compost (COMPOST) and chopped crop residues (RESIDUE), were compared with an unamended control (CTRL). Each amendment supplied the same total amount of carbon, $12.0\,\mathrm{t\,ha^{-1}}$ of carbon over five years, so that differences in retention reflect the material rather than the rate. The treatments were arranged in randomised complete blocks with four blocks at each of the three sites (Gomez and Gomez 1984). The data are held in `data/trials/t5_amendments.tsv`.

5.2.2 Carbon stocks

Soil organic carbon was determined by wet oxidation in topsoil sampled to a depth $d = 0.20\,\mathrm{m}$ at the start and end of the trial. Concentrations were converted to stocks with the bulk density of each site,

$$C = \mathrm{SOC} \times \mathrm{BD} \times d \times 10 \quad (\mathrm{t\,ha^{-1}}), \tag{5.1}$$

with SOC in $\mathrm{g\,kg^{-1}}$, BD in $\mathrm{g\,cm^{-3}}$ and d in metres.

5.2.3 Carbon retention

The change in stock of each plot is $\Delta C = C_{\mathrm{final}} - C_{\mathrm{initial}}$. The fraction of the added carbon retained in the soil is calculated against the control of the same block,

$$f_{\mathrm{ret}} = \frac{\Delta C_{\mathrm{amended}} - \Delta C_{\mathrm{control}}}{C_{\mathrm{added}}}, \tag{5.2}$$

so that the slow change in carbon that occurs without inputs is removed. Mean retention at each site was compared with the clay content of the site, since protection by association with fine mineral particles is expected to increase with clay (Six et al. 2002; von Lützow et al. 2006).

5.3 Results

Only a modest fraction of the added carbon remained in the topsoil after five years (Table 5.1). Averaged over the three amendments, retention was 0.117 at the sandy semi-arid site S2, 0.183 at the temperate site S1 and 0.215 at the clay-rich humid tropical site S3, so that it rose with the clay content of the soil (Figure 5.1).

Table 5.1: Mean fraction of added carbon retained in the topsoil after five years, by site and amendment (four blocks), with the clay content of each site. Sites are ordered by clay content.

Site	Clay (%)	Manure	Compost	Residue	Mean
S2	14	0.105	0.154	0.092	0.117
S1	28	0.187	0.220	0.143	0.183
S3	36	0.214	0.271	0.161	0.215

Compost was retained most efficiently at every site, with fractions of 0.154, 0.220 and 0.271 at S2, S1 and S3. Manure was intermediate (0.105, 0.187 and 0.214), and crop residues were retained least (0.092, 0.143 and 0.161). The ranking of the amendments was the same at all three sites.

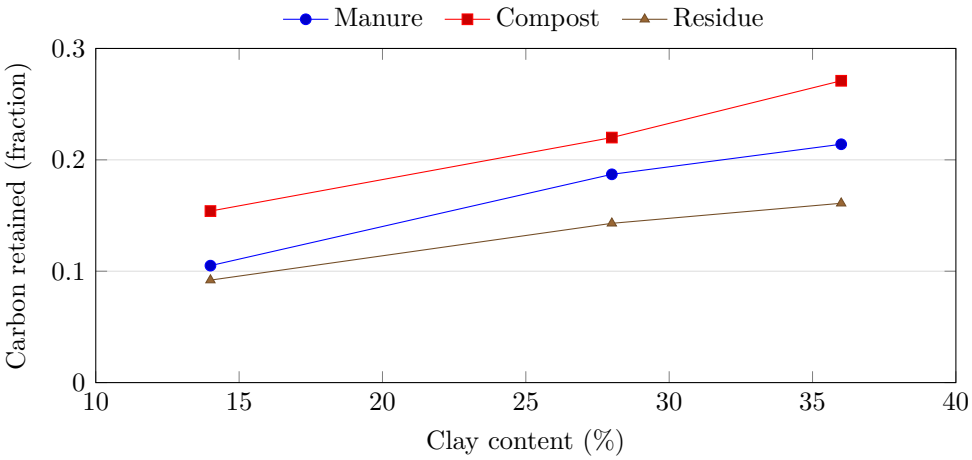


Figure 5.1: Mean fraction of added carbon retained after five years against the clay content of the three sites.

5.4 Discussion

Most of the carbon added in the amendments was lost from the soil within five years, and the fraction retained was small at every site. This agrees with the view that the persistence of organic matter is controlled less by the chemistry of the material added than by the protection it receives once in the soil (von Lützow et al. 2006). The higher retention of compost than of fresh residues is consistent with that view: compost has already lost its most readily decomposed compounds during composting, so a larger share of what remains is available for association with mineral surfaces.

Retention increased with clay content across the three sites. Clay and fine silt provide the surfaces on which organic matter is stabilised and the microaggregates within which it is occluded, so soils with more clay can protect a larger share of each addition (Six et al. 2002). The relationship has a practical implication: the same amendment will build soil carbon more efficiently on a clay soil than on a sandy one, and claims for carbon sequestration from amendments should account for soil texture. With only three sites, however, texture is confounded with climate in this trial, and the semi-arid site is both the sandiest and the driest.

Amendments add carbon that was fixed elsewhere, so their contribution to removing carbon dioxide from the atmosphere is smaller than the gain in soil carbon suggests; part of the benefit is a transfer from one place to another. Their value for soil health and crop nutrition remains, and they form a core component of integrated soil fertility management (Vanlauwe et al. 2010).

5.5 Conclusion

Averaged over three amendments, the fraction of added carbon retained after five years was 0.117, 0.183 and 0.215 at the sites with 14, 28 and 36 per cent clay. Compost was retained best and crop residues least at every site. Texture should be considered whenever the carbon benefit of an amendment is estimated. The data are synthetic and illustrate the calculation only.

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Paper 6

Designing and Analysing Multi-Site Field Trials for Soil Health Research

Contributed paper

Abstract. Field trials in soil health research are usually repeated across sites, because a treatment that works on one soil may fail on another. This paper describes the design shared by the trials reported in this volume: three sites in contrasting climatic zones, randomised complete blocks at each site, and a common set of soil and crop measurements. It sets out how each site is analysed on its own, how the sites are combined, and how the interaction between treatment and site is tested and interpreted. The paper serves as the methodological reference for the other contributions, and all example values are synthetic.

Keywords: field experiments; randomised complete block design; analysis of variance; multi-site trials; treatment by site interaction

6.1 Introduction

Agricultural field experiments must separate the effect of a treatment from the natural variation between plots. The randomised complete block design does so by grouping plots into blocks that are as uniform as possible, assigning every treatment once in each block, and randomising the order of treatments within blocks, so that differences between blocks can be removed from the experimental error (Gomez and Gomez 1984). It remains the most common design for trials on farms and research stations alike.

A single site, however, represents a single soil and climate. When the same experiment is repeated at several sites, the question of interest changes from whether a treatment has an effect to whether its effect is consistent. A treatment by site interaction indicates that the ranking or the size of treatment effects differs between sites, and such interactions are the rule rather than the exception in soil management research, as the variable response of crop yield to reduced tillage across climates illustrates (Pittelkow et al. 2015). Experiments conducted over space and time require their own methods of combined analysis (Gomez and Gomez 1984).

Soil health trials add a further difficulty, because the response is often described not by one variable but by several indicators that must be interpreted together. Converting indicators to a common score before they are combined, as in the Soil Management Assessment Framework (Andrews et al. 2004), makes results comparable across sites, and a clear statement of how indicators were chosen and interpreted is needed if results are to be used by land managers and policy makers (Bünemann et al. 2018).

This paper describes the design and analysis used across the volume. Its aims are to document the layout of the trials, to state the statistical model used at each site and across sites, and to set out the rules by which treatment effects are reported, so that the results of the other papers can be read on a common basis.

6.2 Materials and Methods

6.2.1 Sites

The trials were laid out at three sites chosen to represent contrasting climatic zones: a temperate site (S1), a semi-arid site (S2) and a humid tropical site (S3). The sites differ in soil texture, reaction and organic carbon as well as in climate (Table 6.1). All site and plot data are synthetic and are generated by a seeded program supplied with the volume.

Table 6.1: The three trial sites: clay and sand content, topsoil pH in water, soil organic carbon (SOC) and bulk density (BD) at the start of the trials, mean annual rainfall and mean annual temperature. Values are read from `data/sites.tsv`.

Site	Zone	Clay (%)	Sand (%)	pH	SOC (g kg^{-1})	BD (g cm^{-3})	Rain (mm)	Temp. (°C)
S1	temperate	28	32	6.4	18.0	1.32	780	10.5
S2	semi-arid	14	64	7.6	7.5	1.48	480	24.0
S3	humid tropical	36	40	5.4	14.0	1.25	1450	26.5

6.2.2 Design

Every trial in this volume uses a randomised complete block design with four blocks at each site (Gomez and Gomez 1984). Each block contains every treatment of the trial exactly once, and the order of treatments within each block is randomised independently. Plots are identified by a code giving the trial, site, block and position in the block, and every plot of every trial is listed in a single plot register (`data/trials/plot_register.tsv`), so that each measured value can be traced to its plot.

6.2.3 Analysis at one site

At a single site the measured response y_{ij} of treatment i in block j is described by the model

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \tag{6.1}$$

where μ is the overall mean, τ_i the effect of treatment i, β_j the effect of block j and ε_{ij} the residual error. Treatment effects are tested by analysis of variance, and means are compared by the least significant difference

$$\mathrm{LSD} = t_{\alpha/2,\,\nu}\sqrt{\frac{2\,\mathrm{MSE}}{r}}, \tag{6.2}$$

where MSE is the residual mean square with ν degrees of freedom, r is the number of blocks and $\alpha = 0.05$.

6.2.4 Combined analysis across sites

When the three sites are analysed together, the model is extended to

$$y_{ijk} = \mu + \sigma_k + \beta_{j(k)} + \tau_i + (\sigma\tau)_{ik} + \varepsilon_{ijk}, \tag{6.3}$$

in which σ_k is the effect of site k, $\beta_{j(k)}$ the effect of block j within site k, and $(\sigma\tau)_{ik}$ the interaction between treatment and site (Gomez and Gomez 1984). A significant interaction means that treatment effects differ between sites; results are then reported site by site rather than as a single mean.

6.2.5 Reporting rules

Means are reported with the number of decimal places used in the data files. A treatment effect is described as significant only when the corresponding F-test gives $P < 0.05$, and every table states the number of blocks on which its means are based.

6.3 Results

The combined analysis is illustrated with maize grain yield from the nitrogen rate trial of Paper 2, analysed across the three sites with Equation (6.3) (Table 6.2).

Table 6.2: Combined analysis of variance of maize grain yield in the nitrogen rate trial across the three sites. Sites are tested against blocks within sites; nitrogen and the site by nitrogen interaction against the residual.

Source	df	Sum of squares	Mean square	F	P
site	2	132.82	66.409	228.36	<0.001
block(site)	9	2.62	0.291	3.80	0.002
nitrogen	4	188.34	47.085	615.15	<0.001
site x nitrogen	8	4.64	0.580	7.58	<0.001
error	36	2.76	0.077	–	–

Sites differed strongly ($F = 228.36$, $P < 0.001$), as expected from their contrasting soils and climates, and nitrogen had a large effect ($F = 615.15$, $P < 0.001$). The site by nitrogen interaction was also significant ($F = 7.58$, $P < 0.001$): the size of the response to nitrogen differed between sites, so the response should be reported site by site, as it is in Paper 2, rather than as a single mean. Blocks within sites accounted for a small but significant part of the variation ($F = 3.80$, $P = 0.002$), which confirms that blocking removed real differences in soil fertility from the error.

With 36 residual degrees of freedom, $t_{0.025} = 2.028$, and the least significant difference between two site-by-rate means (Equation (6.2)) is 0.40 t ha^{-1}. Two means that differ by more than this amount differ at the 5 per cent level.

6.4 Discussion

The combined analysis of the nitrogen trial shows why multi-site trials are analysed in two stages. The significant interaction between site and nitrogen means that a single mean response would misrepresent every site, and the results are therefore reported site by site in Paper 2. The same applies to the tillage trial of Paper 4, in which both the size and the direction of the yield response differed between sites. Testing the interaction first, and choosing the level of reporting from its outcome, is the step most often omitted when results from several sites are summarised (Gomez and Gomez 1984).

Three sites are enough to show that treatment effects vary with environment, but not to explain why. Site differs from site in soil, climate and history at once, and with one site per climatic zone those factors cannot be separated. Explaining an interaction requires more sites,

or sites chosen so that one factor varies while the others are held constant, as large syntheses across many trials have done (Pittelkow et al. 2015).

Soil health research adds the further problem of combining several responses. Converting indicators to scores, as in Paper 1, puts them on a common scale, but the choices of scoring limits and weights shape the result and must be reported with it (Andrews et al. 2004; Bünemann et al. 2018). The conventions described here, a common design at every site, one register of plots, analysis of each site before combining, and explicit reporting rules, are intended to make those choices visible and the results reproducible.

6.5 Conclusion

All trials in this volume share a randomised complete block design with four blocks at three sites, a single plot register and a common sequence of analysis. In the nitrogen trial, the site by nitrogen interaction was significant ($P < 0.001$), and two site-by-rate means had to differ by more than 0.40 t ha^{-1} to be distinguished. Multi-site results should be reported by site whenever such an interaction is present. The example values are synthetic.

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List of Abbreviations

Abbreviation	Meaning
AE_N	agronomic efficiency of applied nitrogen
BD	bulk density
CONS	conservation management system
CONV	conventional management system
CT	conventional tillage
INTG	integrated management system
LSD	least significant difference
MBC	microbial biomass carbon
MSE	residual (error) mean square
NT	no-till
P	extractable phosphorus (Mehlich 3)
R0, R1	crop residues removed, retained
RE_N	apparent recovery efficiency of applied nitrogen
SHI	soil health index
SOC	soil organic carbon
WSA	water-stable aggregates

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